Mid-term Exam of Statistics: Principle, Methods and R (I)

8:00-10:30, 6 November 2017

Please read following remarks first!

- The final mark of every one (for the mid-term exam) will be recalibrated on the basis of the points obtained here as follows. Denote the points of student i by X_i, the final mark Y_i for student i is
$$Y_i = \frac{X_i}{\max_j X_j} \times 100.$$
This means that your mark will be recalculated to be the percentage of your points against the highest points among all students.

1. (10pts) Suppose that 30 percent of computer owners use a Macintosh, 50 percent use Windows, and 20 percent use Linux. Suppose that 65 percent of the Mac users have succumbed to a computer virus, 82 percent of the Windows users get the virus, and 50 percent of the Linux users get the virus. We select a person at random and learn that her system was infected with the virus. What is the probability that she is a Windows user?

2. Let U be a Uniform random variable on $[0, 1]$ and define $X = \tan(\pi(U - 1/2))$. Show that the density of X is
$$f_X(x) = \frac{1}{\pi(1 + x^2)}.$$

3. (10pts) Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ and assume that X and Y are independent. Show that the distribution of X given that $X+Y = n$ is $\text{Binomial}(n, \pi)$ where $\pi = \lambda/(\lambda+\mu)$.

4. (10pts) Let
$$f_{X,Y}(x, y) = \begin{cases} c(x + y^3) & 0 \le x \le 1 \text{ and } 0 \le y \le 2 \\ 0 & \text{otherwise.} \end{cases}$$
Find $\mathbb{P}(X < 1/2 \mid Y = 1)$.

5. (10pts)
   (i) (5pts) Let $X_1, \ldots, X_n \sim \text{Poisson}(2)$. Find the limiting distribution of $\overline{X}_n^3 + 2\overline{X}_n^2$.
   (ii) Suppose that $X_n^{(1)}, \cdots, X_n^{(k)}$ are sequences of random variables with $X_n^{(i)} \to 0$ in probablity as $n \to \infty$ for each $i = 1, \cdots, k$.
      (a) (5pts) Show that $\max_{1 \le i \le k} |X_n^{(i)}| \to 0$ in probability as $n \to \infty$.
      (b) (*Extra 5pts*) Find an example to show that the conclusion of (a) is not necessarily true if the number of sequences $k = k_n \to \infty$.

6. (10pts) Let $X \sim \text{Exponential}(\beta)$ with the following density:

$$f_X(x) = \begin{cases} \beta e^{-\beta x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

Find $\mathbb{P}(|X - \mu_X| \geq t\sigma_X)$ for $t > 1$, where μ_X and σ_X^2 are the mean and the variance of X. Compare this to the bound you get from Chebyshev's inequality.

7. (10pts) Suppose (X, Y) are mean zero, variance one random variables with $\text{Cov}(X, Y) = \rho$.
   (a) Find the constant a that minimizes $\mathbb{E}(Y - aX)^2$. Note that the same a applies to the rest of the problem.
   (b) If (X_0, Y_0) has the same joint distribution as (X, Y), but is independent of (X, Y), find the mean and variance of $aX + (Y_0 - aX_0)$.
   (c) Find the covariance between X and $aX + (Y_0 - aX_0)$.
   (d) If (X, Y) is additionally distributed as a bivariate normal, what can you say about the joint distribution of $(X, aX + (Y_0 - aX_0))$ as compared to (X, Y)?
   (e) Suppose (X, Y) are jointly normal and you are given an observation $X = x$. How would you create a new observation y such that (x, y) was a realization from the distribution of (X, Y)?

8. (20pts) Read the following R code.

```r
### first part ###
monte.carlo <- function(n)
  # n is the sample size
{
  x <- rnorm(n, mean = 3, sd = 1)
  h <- exp(cos(x))/sqrt(2*pi)
  list(integral=mean(h), SE=sqrt(var(h)/n), SampleSize=n)
}

results <- data.frame( integral = numeric(0),
                       SE = numeric(0), SampleSize = integer(0))

for (i in 1:100)
{
   results <- rbind(results, monte.carlo(1000*i))
}

plot(results$SampleSize, results1$integral,
     xlab = "Sample Size", ylab = "Integral Value", type = "o")
```

```r
### second part ###

par(mar=c(3,3,3,2),mfrow=c(2,2))

no.seq = c(100, 200, 500, 1000)
# no.seq = c(20, 30)
for (i in 1:length(no.seq))
{
  no <- no.seq[i]
  # reset results first
  results <- data.frame( integral = numeric(0),
                         SE = numeric(0), SampleSize = integer(0))
  for (j in 1:1000)
  {
     results <- rbind(results, monte.carlo(no))
  }
  hist(results$integral, breaks = 20, probability=T,
       main=sprintf('n=%d',no), xlab="Integral")
}
```

The R function `rbind` puts extra records in the data frame and works as below

```r
> n = c(2, 3, 5)
> b = c(TRUE, FALSE, TRUE)
> df <- data.frame(n, b)
> df
  n     b
1 2  TRUE
2 3 FALSE
3 5  TRUE
> df <- rbind(df, list(7, FALSE))
> df
  n     b
1 2  TRUE
2 3 FALSE
3 5  TRUE
4 1 FALSE
```

(a) Please explain in words and/or formulas what the function `monte.carlo` is doing.

(b) After running the first part of the code, we obtain Figure 1. Explain what you see and find the mathematical theorem responsible for what you see.

(c) After running the second part of the code, we obtain Figure 2. Compare the four subplots, explain what you see and find the mathematical theorem to responsible for what you see.

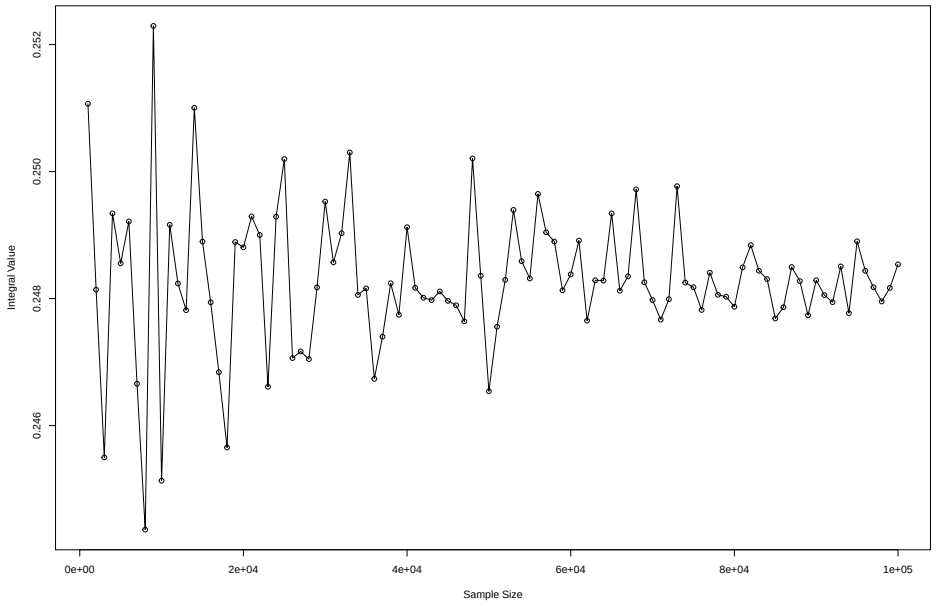


Figure 1: Sample Mean Chart

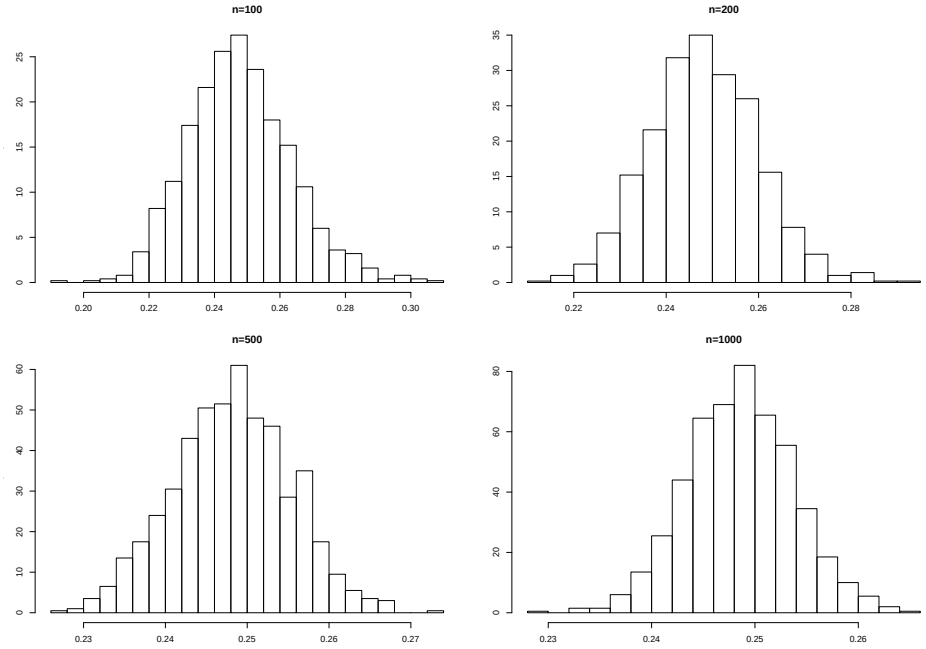


Figure 2: Histogram Plots